

Title: CUSTOMER MASK GENERATION GUIDE

1. PURPOSE/SCOPE

1.1 Purpose

The purpose of this document is to give SkyWater customers a general overview of the procedures that occur during tapeout of their design submission. This document will illustrate with examples the various steps that occur from design submission to generated mask data sent to the mask vendor. These examples are intended to aid in the customer's understanding of each step and how features are being manipulated. It is also intended to give the customer a good estimate of the turn time of the process. A process map of this flow is illustrated in Appendix A of this document.

1.2 Scope

This document is applicable to all tapeout submissions where SkyWater builds the reticle frame.

2. RESPONSIBILITIES

2.1 The Design Enablement Director owns this document.

2.2 The Tapeout Engineer is responsible for following this document and keeping the document up to date with any changes in procedures.

2.3 Customer Managers are responsible for communicating this document to SkyWater customers.

3. REFERENCED DOCUMENTS

3.1 [Referenced Documents \(DMS\)](#)

CTRL-Click (click if PDF) on the above link to access the Document Management System (DMS) details page for the spec. Then click on the "Reference Documents"  button to bring up the list of referenced documents.

4. MATERIALS: N/A

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5. EQUIPMENT: N/A

6. SAFETY: N/A

7. CRITICAL REQUIREMENTS SUMMARY: N/A

8. OPERATING PROCEDURES AND RESPONSIBILITIES

8.1 GDS Design Submission to SkyWater

Designs are submitted in GDSII format to SkyWater's secure FTP (compressed in .gz format). Customers should request an FTP account if one has not been setup before submission.

Once the design has been received by SkyWater a Tapeout Engineer will be assigned to the job and the assigned Technical Lead or Customer Manager will be contact with the customer on the status of the job.

8.2 Verification Checks

8.2.1 Manufacturing Rule (MR) Checks

These are rule checks needed for manufacturability. MR checks are usually minimum line and space rules and critical surround rules. Any MR violation found in a design must be fixed. These checks are denoted with an "MR" in the Technology Family Design Manual.

8.2.2 Density Checks

SkyWater requires that customers meet pattern density requirements on layers such as active area, poly and metals.

8.2.2.1 Full Area Density Checks

There are global density requirements on some layers for the die. These requirements are so that the frame image on the reticle can be within a certain range of density to allow for consistent etching. These rules can be found in the Technology Family Design Manual.

8.2.2.2 CMP Pattern Density Checks

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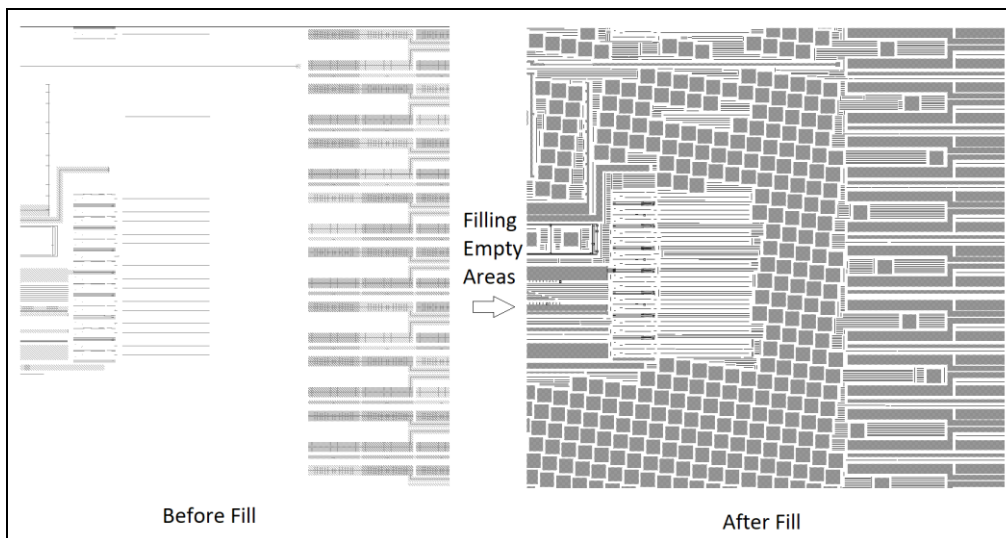
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These checks are a stepped window check (700 x 700 window size stepped 350 microns) to assure that density is met across the design. The purpose of this is to avoid adverse polishing effects during CMP processing due to very sparse and/or very dense areas.

8.2.2.3 Filling Layers

Typically layers such as active area, poly and metals must have empty areas filled with a consistent fill pattern. SkyWater prefers that customers do their own pattern fill in empty areas to meet density requirements.

Met1 Fill Example:



8.3 Build Frame

This step involves generating an array of dice, whether it be a single die or a MPW (multi-project wafer). SkyWater may generate frames for customer submitted products. This involves placing e-test and fab block structures in the scribe lines that are used to monitor the wafer process.

8.4 Created Layer Design Rule Check (CLDRC)

CLDRC is the process, post frame build, that creates mask layers, OPC and checks pattern density rules. It outputs a gds file that is ready to send to the mask vendor if all rules and requirements are met.

8.4.1 Created Layers

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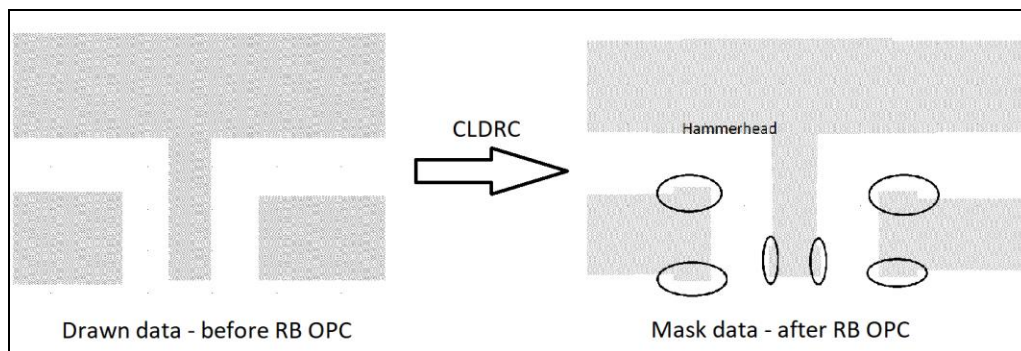
This step takes the drawn data in the frame and converts them to mask data via Boolean and sizing operations. Keep in mind that for some of the implant layer definitions there are drawn layers inside the equations that are not defined in the document due to the complexity. Note that drawn layer names may differ from mask layer names.

8.4.2 OPC (Optical Proximity Correction)

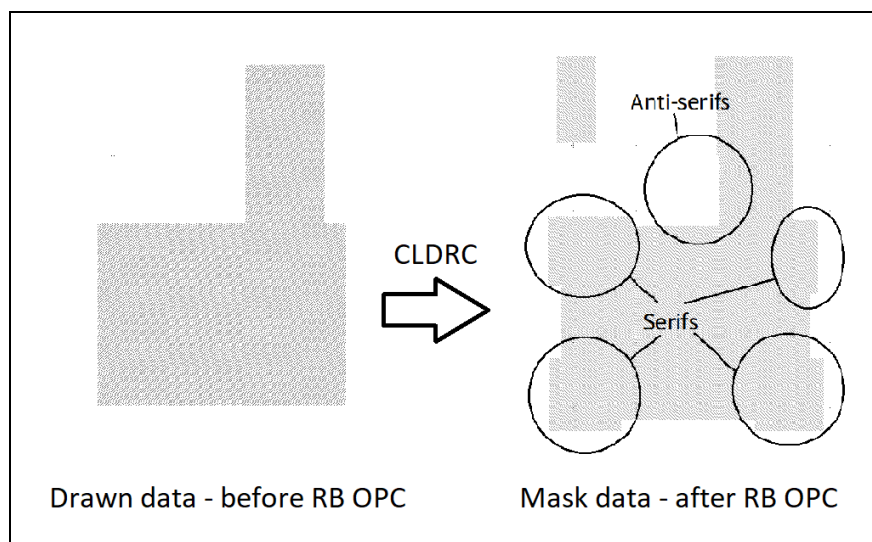
8.4.2.1 Rule Based OPC

This step adds features to the drawn data on certain layers. Features such as hammerheads, serifs/anti-serifs and line lengthening. Below are examples illustrating these types of rule based (RB) OPC modifications.

Example: Hammerheads



Example: Serifs and Anti-serifs

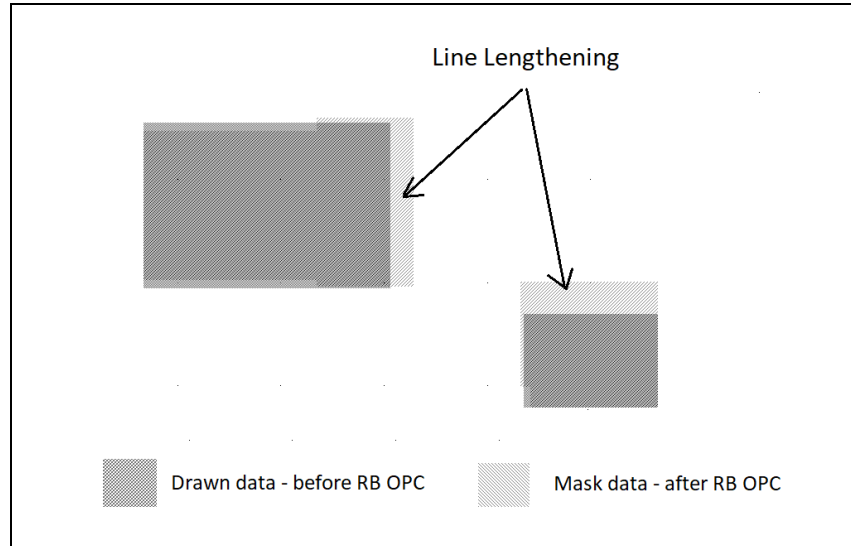


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Example: Line Lengthening



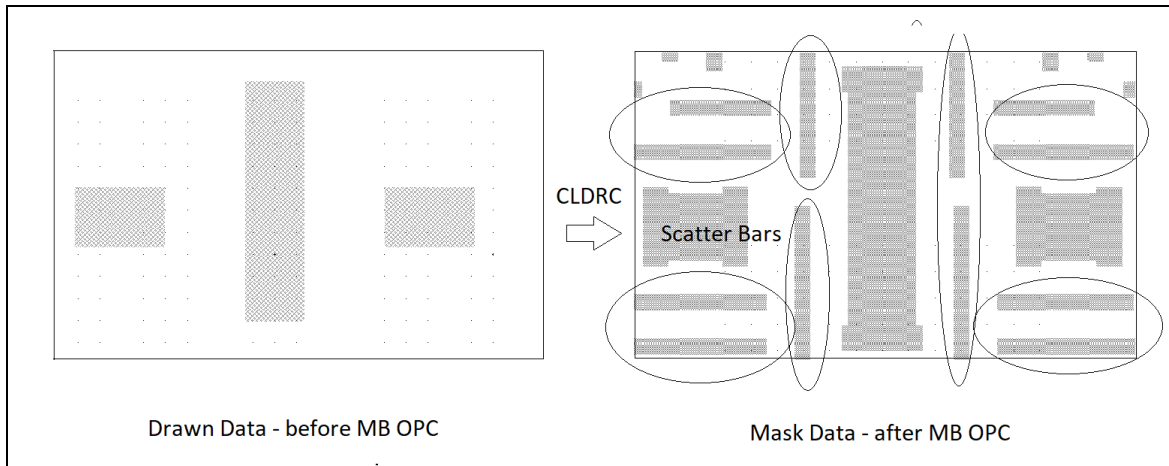
8.4.2.2 Model Based OPC

Model Based (MB) OPC is done using models based on optical characteristics of the lithography tools and photoresist. SkyWater's OPC software tool uses these models to determine what modifications are done on the drawn features so that they print correctly on the wafer.

8.4.2.2.1 Scatter Bars

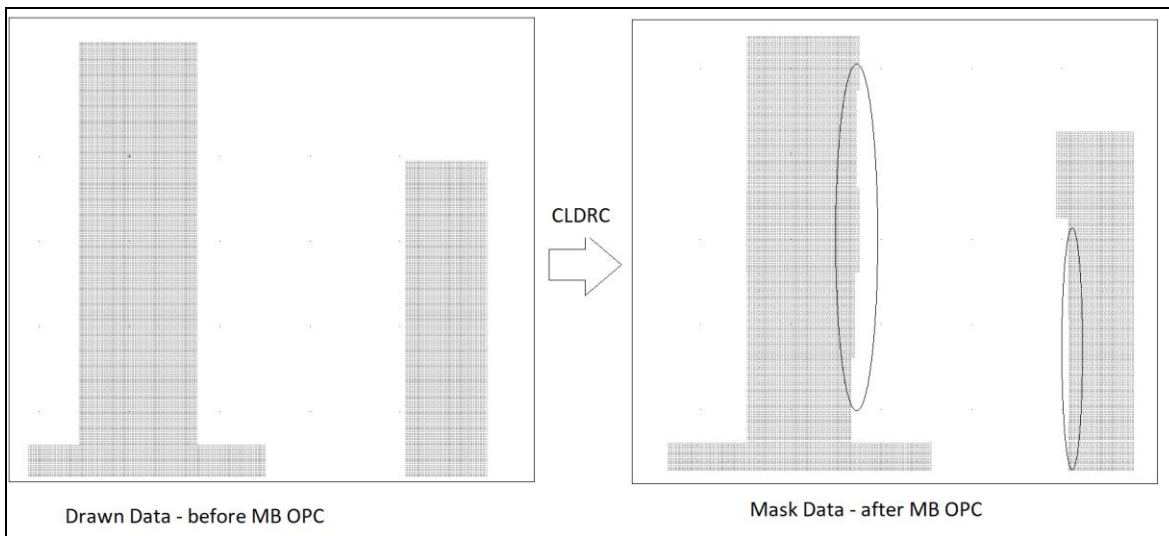
Scatter bars, also known as “assist features”, are placed around lines of drawn data to create an equivalent dense environment. This modification improves image quality of the drawn feature they are surrounding. Scatter bars themselves do not print on the wafer as they are too small to resolve – they are typically less than the minimum design rule width.

Example: Scatter Bars (Assisting Feature)



8.4.2.2.2 Line Biasing

Edges of lines are adjusted during MB OPC. The OPC software analyzes the surrounding features to these edges when adjusting line width.



8.4.3 Percent Clear

A percent clear calculation is needed for each mask ordered. It is the percentage of the area on the mask that is not chrome (or clear area on mask). Etch and implant engineers use these values for processing. A report is generated for each tapeout, it contains percent clear values for each layer.

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8.4.4 TSAC (Total Surface Area Calculation)

TSAC values are calculated on active area, poly and interconnect layers. There is a report generated from the CLDRC log file. CVD engineers use these numbers for processing.

8.5 Submitting Mask Order

After the CLDRC step a gds file of the mask data is sent to the mask vendor once the mask job has been approved (internally at SkyWater). This is done once there has been an internal tapeout review, all paperwork has been generated and approved and all checks have been completed.

8.6 Jobdeck Review and Approval

The mask vendor will generate a “jobdeck” of the mask order. This jobdeck will contain the final form of the data before it is written on the plate. Before any plates in the job can be written the mask vendor will need approval from both SkyWater and the customer. Jobdecks are typically ready within 24 hours of the mask order submission.

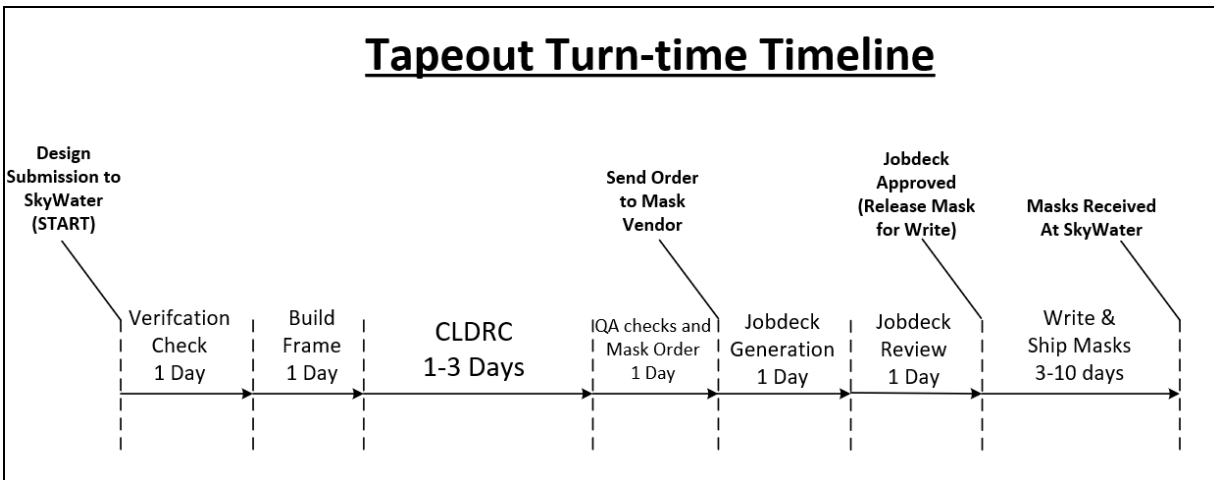
IMPORTANT NOTE: When viewing jobdecks keep in mind that the presence of data represents clear area on the mask.

8.7 Writing and Shipping of Masks

Once the mask vendor has received approval from both the customer and SkyWater then masks will be written. The turn time of the masks being written will depend on mask grade, mask vendor queue, job priority and shipping priority.

8.8 Tapeout Turn-time

The timeline for tapeouts is dependent on the steps illustrated in the process map in Appendix A. Below is an illustration of the tapeout timeline and an estimate of how long each step should take. Keep in mind that any step in the illustrated timeline may take longer if all requirements are not met or unexpected issues arise. When errors do arise that need correction it will be necessary in many cases to cycle back to the start of the timeline and repeat the process.



9. QUALITY REQUIREMENTS: N/A

10. RECORDS

10.1 Storage location and retention period for Document Control records is specified in procedure 00-00064 – Record Retention Policy and Procedures

10.2 Document History

[Document History Query](#)

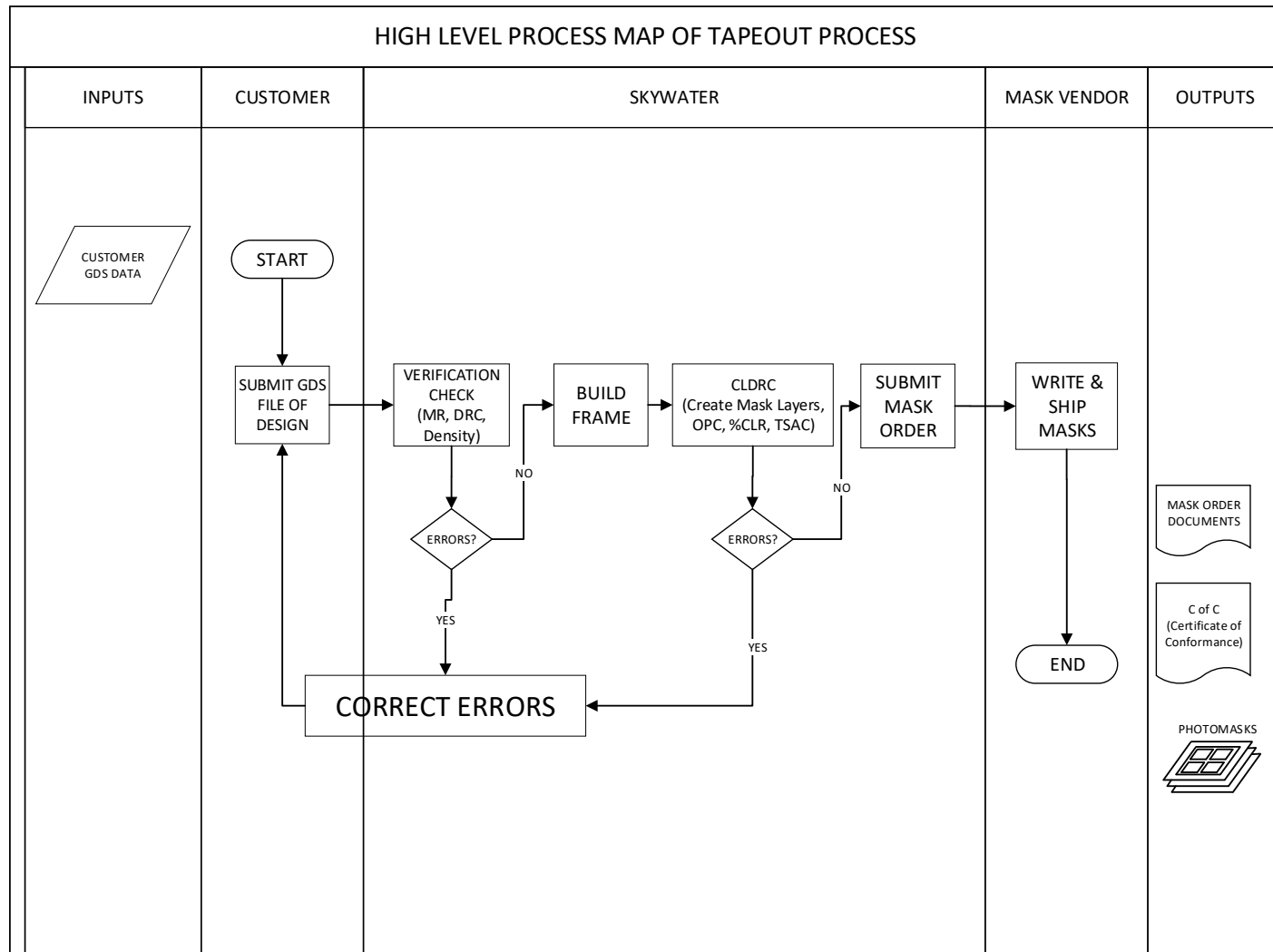
To access the document revision history, CTRL-click (click if PDF document) on the above link to bring up the web query page and enter the spec number in the dialog box and click on submit.

11. PREVENTIVE MAINTENANCE: N/A

12. POSTING SHEETS/FORMS/APPENDIX

12.1 Appendix A – TAPEOUT PROCESS MAP

Appendix A – TAPEOUT PROCESS MAP



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